CAMPUS NETWORK DESIGN & IMPLEMENTATION

1. Executive Summary & Architecture Overview

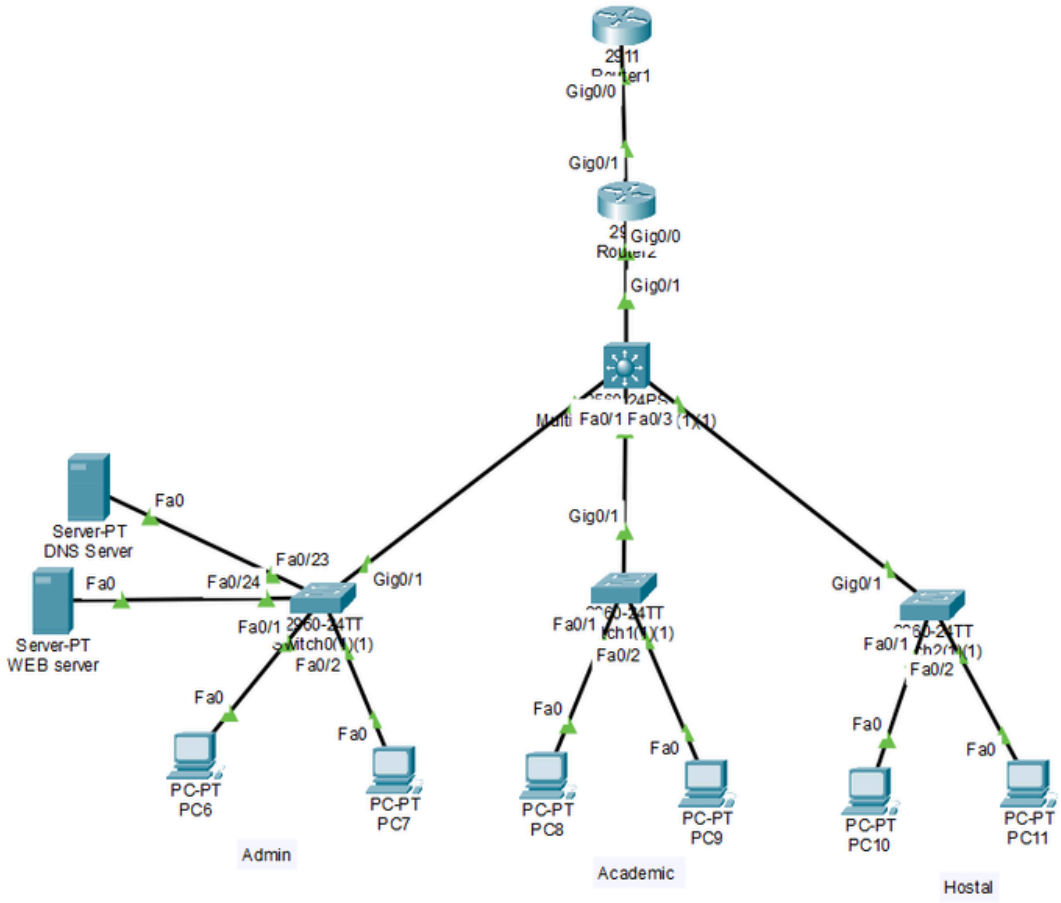
This technical document details the engineering design, VLSM subnetting strategy, multilayer switching architecture, routing protocols, and advanced security configurations deployed for the campus network at Cosmos College [cite: 1]. The architecture implements a structured hierarchical topology encompassing Administration, Academic, and Hostel blocks, paired with a protected Server Farm and redundant wide-area network (WAN) gateways.

Key Highlights of Implementation:

- **VLAN Segmentation:** Complete Layer 2 isolation across 7 distinct VLANs (Admin, Academic, Hostel, and Server blocks).
- **Inter-VLAN Routing:** Layer 3 SVI (Switched Virtual Interface) configuration on the Core Multilayer Switch.
- **Security Policies:** Strict Port Security (sticky MAC address learning with restricted violations) and Extended Access Control Lists (Hostel Block policy).
- **WAN Connectivity:** Multi-router setup featuring dynamic routing fallback/static routing configurations.

2. Network Topology

The visual architecture below displays the physical and logical placement of the core multilayer switch, departmental access switches, end-user nodes, servers, and perimeter routers within Cisco Packet Tracer.



3. VLSM IP Addressing Scheme

Variable Length Subnet Masking (VLSM) was meticulously implemented to optimize address space allocation across the institutional departments without wastage.

Department / Zone	VLAN ID	Subnet / CIDR	Subnet Mask	Gateway IP
Admin Dept 1	10	192.168.10.0/26	255.255.255.192	192.168.10.1
Admin Dept 2	20	192.168.10.64/26	255.255.255.192	192.168.10.65
Academic Dept 1	30	192.168.10.128/27	255.255.255.224	192.168.10.129
Academic Dept 2	40	192.168.10.160/27	255.255.255.224	192.168.10.161
Hostel Dept 1	50	192.168.10.192/28	255.255.255.240	192.168.10.193
Hostel Dept 2	60	192.168.10.208/28	255.255.255.240	192.168.10.209
Server Farm WAN	99	192.168.10.224/28	255.255.255.240	192.168.10.225
Link	-	192.168.10.232/30	255.255.255.252	192.168.10.234

4. Complete Device Configurations

4.1 Core Multilayer Switch Configuration

```
enable
configure terminal
ip routing

vlan 10
name Admin-Dept1
vlan 20
name Admin-Dept2
vlan 30
name Academic-Dept1
vlan 40
name Academic-Dept2
vlan 50
name Hostel-Dept1
vlan 60
name Hostel-Dept2
vlan 99
name Servers
exit

interface vlan 10
ip address 192.168.10.1 255.255.255.192
 no shutdown
```

```
exit

interface vlan 20
ip address 192.168.10.65 255.255.255.192 no shutdown
exit

interface vlan 30
ip address 192.168.10.129 255.255.255.224 no shutdown
exit

interface vlan 40
ip address 192.168.10.161 255.255.255.224 no shutdown
exit

interface vlan 50
ip address 192.168.10.193 255.255.255.240 no shutdown
exit

interface vlan 60
ip address 192.168.10.209 255.255.255.240 no shutdown
exit

interface vlan 99
ip address 192.168.10.225 255.255.255.240 no shutdown
exit

interface range fa0/1 - 3
switchport trunk encapsulation dot1q switchport mode trunk
no shutdown
exit

interface gigabitethernet0/1

 no switchport
 ip address 192.168.10.234 255.255.255.252
 no shutdown
 exit
 ip route 0.0.0.0 0.0.0.0 192.168.10.233

 ip access-list extended HOSTEL_POLICY
 deny ip 192.168.10.192 0.0.0.15 192.168.10.224 0.0.0.15
 deny ip 192.168.10.208 0.0.0.15 192.168.10.224 0.0.0.15
 permit tcp 192.168.10.192 0.0.0.15 host 192.168.10.230 eq www
 permit tcp 192.168.10.208 0.0.0.15 host 192.168.10.230 eq www
 permit udp 192.168.10.192 0.0.0.15 any eq domain
 permit udp 192.168.10.208 0.0.0.15 any eq domain
 permit ip 192.168.10.192 0.0.0.15 any
 permit ip 192.168.10.208 0.0.0.15 any
 exit

 interface vlan 50
 ip access-group HOSTEL_POLICY in
 exit

 interface vlan 60
 ip access-group HOSTEL_POLICY in
 exit
```

4.2 Perimeter Router 1 Configuration

```
enable
configure terminal
interface GigabitEthernet0/0
ip address 10.0.0.1 255.255.255.252
no shutdown
exit

interface GigabitEthernet0/1
ip address 192.168.10.233 255.255.255.252
no shutdown
exit

ip route 192.168.10.0 255.255.255.0 192.168.10.234
ip route 0.0.0.0 0.0.0.0 10.0.0.2
exit
```

4.3 External Router 0 Configuration

```
enable
configure terminal
interface GigabitEthernet0/0
ip address 10.0.0.2 255.255.255.252
no shutdown
exit

ip route 192.168.10.0 255.255.255.0 10.0.0.1
exit
```

4.4 Access Switches Configuration

Admin Switch 0 Configuration:

```
enable
configure terminal
vlan 10
vlan 20
vlan 99
exit

interface gigabitethernet0/1
switchport mode trunk
no shutdown
exit

interface fa0/1
switchport mode access
switchport access vlan 10
switchport port-security
switchport port-security maximum 2
switchport port-security violation restrict
 switchport port-security mac-address sticky
 no shutdown
exit

interface fa0/2
switchport mode access
switchport access vlan 20
switchport port-security
switchport port-security maximum 2
 switchport port-security violation restrict
switchport port-security mac-address sticky
no shutdown
exit

interface range fa0/23 - 24
switchport mode access
switchport access vlan 99
no shutdown
exit
```

Academic Switch 1 Configuration:

```
enable
configure terminal
vlan 30
vlan 40
exit
interface gigabitethernet0/1
switchport mode trunk
no shutdown
exit
interface fa0/1
switchport mode access
switchport access vlan 30
switchport port-security
switchport port-security maximum 2
 switchport port-security violation restrict
switchport port-security mac-address sticky no shutdown
exit
enable
configure terminal
vlan 30
vlan40
```

```
exit
interface gigabitethernet0/1
switchport mode trunk
no shutdown
exit
interface fa0/1
switchport mode access
switchport access vlan 30
switchport port-security
switchport port-security maximum 2
switchport port-security violation restrict
 switchport port-security mac-address sticky no shutdown
exit
interface fa0/2
switchport mode access
switchport access vlan 40
switchport port-security
switchport port-security maximum 2 switchport port-security violation restrict
switchport port-security mac-address sticky no shutdown
exit
```

Hostel Switch 2 Configuration:

```
enable
configure terminal
vlan 50
vlan 60
exit

interface gigabitethernet0/1
switchport mode trunk
no shutdown
exit

interface fa0/1
switchport mode access
switchport access vlan 50
switchport port-security
switchport port-security maximum 2
switchport port-security violation restrict
switchport port-security mac-address sticky no shutdown
exit

interface fa0/2
switchport mode access
switchport access vlan 60
switchport port-security
switchport port-security maximum 2
switchport port-security violation restrict
switchport port-security mac-address sticky
no shutdown
exit
```

```
enable
configure terminal
vlan 50
vlan 60
exit

interface gigabitethernet0/1
switchport mode trunk
no shutdown
exit

interface fa0/1
switchport mode access
switchport access vlan 50
switchport port-security
switchport port-security maximum 2
switchport port-security violation restrict
switchport port-security mac-address sticky no
shutdown
exit

interface fa0/2
switchport mode access
switchport access vlan 60
switchport port-security
switchport port-security maximum 2
switchport port-security violation restrict
switchport port-security mac-address sticky no
shutdown
exit
```

5. Security Policy Implementation Details

To satisfy campus security requirements, two primary mechanisms were enforced:

- **Port Security:** Configured on access switch ports (`fa0/1` and `fa0/2`) across all departmental switches. Maximum allowed MAC addresses per port is restricted to 2, using sticky learning with a restrict violation mode to prevent unauthorized device plug-ins.
- **Extended Access Control List (HOSTEL_POLICY):** Applied inbound on Hostel VLAN interfaces (VLAN 50 and 60). This policy restricts hostel users from directly accessing the internal server farm network except for permitted HTTP web services (port 80) and DNS resolution, while permitting open external/internet traversal.

6. System Verification & Testing

Connectivity validation was verified via extensive testing, including intra-departmental pings, inter-VLAN pings routed via the core multilayer switch, web server HTTP validation, and ACL blockage compliance tests.

